Weeks 3–4 | **In-Class Quiz I** week 4 | **Lab Quiz I** week 5 | **Midterm** 3/11/2026 | **File** 1 / 8

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Chapter 3 exam brief

ماذا يطلب الكورس أوت لاين من الفصل الثالث — اقرأ هذا أولًا

This set merges two lecture decks into one: **Dr. Amani Al-Rubkhi's Fall 2026 notes** and **Dr. Ahmed Al-Hadhrami's Spring 2024 slides**. Every definition, rule, example and activity from both appears exactly once. Where both decks teach the same thing, it is written once and tagged with both sources. Dr. Amani is the **course coordinator** named in the outline, so where the two decks differ, her version comes first.

CHAPTER 3 IN THE FALL 2026 SCHEDULE

WEEK	LECTURES	LAB (R)	ASSESSMENT / EXERCISES
3	3-1 Measures of Central Tendency 3-2 Measures of Variation	Lab 3 — RStat2101: Data Basics (cont.)	Exercises Chapter 3, **page 178: {2, 3, 10, 17}** + Moodle exercises
4	3-3 Measures of Position 3-4 Exploratory Data Analysis	Lab 4 — RStat2101: Tabulation, Graphs	Moodle exercises (cont.) + **In-Class Quiz I**
5	(Chapter 4 starts)	Lab 5 — RStat2101: **Descriptive Measures**	**Lab Quiz I** (R)
8	—		**Midterm** 25% · 3/11/2026 · Chapters 1, 2, 3, 4
16	—		**Final** 45% · all chapters

The outcomes Chapter 3 is graded against

OUTCOME	CRITERION (VERBATIM FROM THE OUTLINE)	ASSESSED BY
CLO1-A	Define essential statistical terminology, including … sample, population, **parameters, and statistics**.	Tutorial, In-Class Quiz, Midterm, Final
CLO2-A	**Calculate** different statistical measures including measures of central tendency, measures of variability and measures of position.	Tutorial, In-Class Quiz, Active Learning, Midterm, Final
CLO2-D	Use R to perform basic descriptive statistics, create tables, and visualize data.	Active Learning, **Lab Quiz**

OUTCOME	CRITERION (VERBATIM FROM THE OUTLINE)	ASSESSED BY
CLO3-B	Extract meaningful insights from visual data representations.	Tutorial, In-Class Quiz, Midterm
CLO3-C	**Derive reasoned insights** based on the interpretation of graphs and summary statistics.	Tutorial, In-Class Quiz, Midterm
CLO3-D	Utilize R to perform exploratory data analysis.	Active Learning, Lab Quiz

WHAT THIS MEANS FOR HOW YOU STUDY

Two of the outcomes are about **interpretation** (CLO3-B, CLO3-C), not calculation. Exam questions in this chapter rarely stop at "find the mean". They continue: *which measure is more appropriate, and why?* · *which data set is more consistent?* · *what does this boxplot say about the shape?* · *what can be said about the number of observations in this interval?* Every calculation in these files ends with a sentence of interpretation. Write that sentence in the exam too, because it carries marks.

الخلاصة بالعربي

الفصل الثالث يُدرَّس في الأسبوعين **3 و 4**، ويُختبر في **الكويز الصفي الأول** (الأسبوع 4)، ثم في **كويز اللاب الأول** بلغة R (الأسبوع 5)، ثم في الميدتيرم والنهائي. المخرجات المطلوبة نوعان: **الحساب** (CLO2-A: مقاييس النزعة المركزية والتشتت والموقع)، و**التفسير** (CLO3-B و CLO3-C: استخراج استنتاجات من الرسوم والمقاييس). لذلك لا يكفي أن تحسب الرقم، بل يجب أن تكتب ماذا يعني. التمارين المطلوبة من الكتاب: صفحة 178، الأسئلة **2 و 3 و 10 و 17**.

Where the two decks differ, and what to write in the exam

#	POINT	WHAT TO DO
1	Dr. Amani's population-variance table labels `N` as "sample size".	It is the **population size**. `N` always means population and `n` means sample.
2	Her population SD box is headed "Population variance, σ".	σ is the population **standard deviation**; σ² is the variance.
3	"Coefficient of variance" in one activity.	The term is **coefficient of variation** (CVar).
4	CV use. Dr. Ahmed: "same means but different SDs". Dr. Amani: "different means and different SDs".	Both are true. CV works in every case and is **essential** when the means or the units differ.
5	Dr. Amani's variance example for 1, 2, 4 gives `s² = 2.335`.	The exact value is `7/3 = 2.333`. The 2.335 comes from rounding deviations early, which is exactly what the general rounding rule warns against (see 3-2).

#	POINT	WHAT TO DO
6	Homework stem-and-leaf: printed answers (mean 19.6, median 17) do not match the plotted 14 values.	Both versions are solved in 3-1. The printed answers need one more value, 35 (a leaf "3 | 5"), that is missing from the plot.
7	Car-accident activity: data 22, 30, 36, 24, 5 with the note "mean = 28".	Those values have mean 23.4. The note fits only if the last value is 28. Both versions are solved in 3-2.
8	Median of grouped data appears only in Dr. Amani's notes, and only as "find the median class".	Give the **median class**. A one-number interpolation is shown as optional, only if a question asks for a value.
9	Z-scores, the empirical rule, deciles, midrange and linear transformations appear only in Dr. Ahmed's slides.	All are included. They are in the textbook (Bluman 10e) that the outline names, so treat them as examinable.

The method rule for quartiles and percentiles

Hand methods, R and calculators can give different quartiles for the same data. On paper, use the course's procedure tables. In 3-3 and 3-4 you will see where R's `summary()` and `quantile()` disagree with them. If a question names a method ("using the percentile formula"), use that method.

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · 01 · Lesson 3-1: Measures of Central Tendency · 02 · Lesson 3-2: Measures of Variation · 03 · Lesson 3-3: Measures of Position · 04 · Lesson 3-4: Exploratory Data Analysis · 05 · Revision pack: formulas, decisions, traps, R · 06 · Exam-style questions (no answers) · 07 · Full worked answers